# Homework 11

## This homework is due at 11 PM on April 17, 2026.

**Submission Format:** Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned), as well as a printout of your completed Jupyter notebook(s).

1. **Linear Programming**

   Express the following problem as an LP.

$$\min_{\vec{x} \in \mathbb{R}^k} \quad \sum_{i=1}^{k} |x_i| \tag{1}$$

$$\text{s.t.} \quad A\vec{x} = \vec{b} \tag{2}$$

**2. Fun with Hyperplanes**

In this problem we work with hyperplanes, which are key components of linear programming as well as future topics such as support vector machines.

(a) Sketch the hyperplane $\mathcal{H} \doteq \{\vec{x} \in \mathbb{R}^2 \mid \begin{bmatrix} 1 & 1 \end{bmatrix} \vec{x} = 2\}$.

(b) Let $\vec{c} \in \mathbb{R}^n$ be nonzero, and let $\mathcal{H} \doteq \{\vec{x} \in \mathbb{R}^n \mid \vec{c}^\top \vec{x} = 0\}$. Show that $\mathcal{H}$ is a linear subspace of $\mathbb{R}^n$. What is $\dim(\mathcal{H})$?

(c) Let $\vec{c} \in \mathbb{R}^n$ be nonzero, and let $\mathcal{H} \doteq \{\vec{x} \in \mathbb{R}^n \mid \vec{c}^\top \vec{x} = 0\}$. Suppose $\vec{x}_\star \in \mathbb{R}^n$ is on one side of the hyperplane, i.e., $\vec{c}^\top \vec{x}_\star > 0$. Give any vector which is on the other side of the hyperplane but not on the hyperplane itself.

(d) Let $\vec{c} \in \mathbb{R}^n$ be nonzero, and let $\vec{x}_0 \in \mathbb{R}^n$ be arbitrary. Let $\mathcal{H} \doteq \{\vec{x} \in \mathbb{R}^n \mid \vec{c}^\top (\vec{x} - \vec{x}_0) = 0\}$. Suppose $\vec{x}_\star \in \mathbb{R}^n$ is on one side of the hyperplane. Give any vector which is on the other side of the hyperplane but not on the hyperplane itself.

(e) Let $\vec{x}_0 \in \mathbb{R}^n$ be arbitrary. For a vector $\vec{c} \in \mathbb{R}^n$, let $\mathcal{H}(\vec{c}) \doteq \{\vec{x} \in \mathbb{R}^n \mid \vec{c}^\top (\vec{x} - \vec{x}_0) = 0\}$. Show that $\vec{0} \in \mathcal{H}(\vec{c})$ for every $\vec{c} \in \mathbb{R}^n$ if and only if $\vec{x}_0 = \vec{0}$.

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**3. LP at Boundary**

Consider the LP:

$$\min_{x \in \mathbb{R}^n} \quad \vec{c}^\top \vec{x} \tag{3}$$

$$\text{s.t.} \quad A\vec{x} \leq \vec{b}. \tag{4}$$

Here we have non-zero $\vec{c} \in \mathbb{R}^n$, $\vec{x} \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, and $\vec{b} \in \mathbb{R}^m$. The feasible set forms a polyhedron

$$\mathcal{P} = \{\vec{x} \in \mathbb{R}^n \mid \vec{a}_i^\top \vec{x} - b_i \leq 0,\ 1 \leq i \leq n\} \tag{5}$$

$$= \bigcap_{i=1}^{m} \{\vec{x} \in \mathbb{R}^n \mid \vec{a}_i^\top \vec{x} - b_i \leq 0\}, \tag{6}$$

where $\vec{a}_1^\top, \ldots, \vec{a}_n^\top$ are the row vectors of $A$. In the second line, $\mathcal{P}$ is defined as the intersection of half-spaces. When this set is bounded, it is often referred to as a polytope instead.

In this problem we will prove facts about linear programs, including the crucial fact that the solution to an LP exists at the boundary of a polytope.

(a) First, consider the case where we have an unbounded polyhedron. For any such $\mathcal{P}$, assume there exists some $\vec{x}_0, \vec{v} \in \mathbb{R}^n$ with $\vec{x}_0 \neq \vec{v}$ such that $\mathcal{L} := \{\vec{x}_0 + \alpha\vec{v} \mid \alpha \in [0, \infty)\} \subset \mathcal{P}$. This is saying that there exists a line segment, unbounded on one side, that is contained within $\mathcal{P}$. Show that if $\vec{c}^\top \vec{v} < 0$, then $p^* = -\infty$.

(b) Now, suppose our feasible set is defined by a bounded polytope (i.e. a polytope that contains its boundary). We say a point belongs to the interior of $\mathcal{P}$ (i.e. $\vec{x}_0 \in \text{Int}(\mathcal{P})$) if there exists some ball of radius $\epsilon > 0$ such that $\mathcal{B} := \{\vec{y} \in \mathbb{R}^n \mid \|\vec{y} - \vec{x}_0\|_2 \leq \epsilon\} \subset \mathcal{P}$. Show that the optimal point for the LP cannot be obtained in the interior of $\mathcal{P}$ and must be obtained on the boundary (that is, when for some $1 \leq i \leq n$, $\vec{a}_i^\top \vec{x} - b_i = 0$). For this, consider a proof by contradiction. Show that for any $x_0$ on the interior, there exists another point

$$\vec{x}_1 := \vec{x}_0 - \epsilon \frac{\vec{c}}{\|\vec{c}\|_2}, \tag{7}$$

such that $\vec{x}_1 \in \mathcal{P}$ and $\vec{c}^\top \vec{x}_1 < \vec{c}^\top \vec{x}_0$.

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**4. Linear Programs**

Consider the following linear program:

$$\max_{x_1, x_2 \in \mathbb{R}} \quad 2x_1 + 3x_2 \tag{8}$$

$$\text{s.t.} \quad x_1 + 2x_2 \leq 8 \tag{9}$$

$$x_1 - x_2 \leq 2 \tag{10}$$

$$x_2 + x_1 \geq 2 \tag{11}$$

$$x_1 \geq 0 \tag{12}$$

$$x_2 \geq 0 \tag{13}$$

(a) Sketch the feasible region of the linear program as well as the 5-, 10-, and 15-level sets of the objective function.

*HINT: Recall that for $f : \mathbb{R}^n \mapsto \mathbb{R}$ and $\alpha \in \mathbb{R}$, the $\alpha$-level set of $f$ is defined as $\{\vec{x} \in \mathbb{R}^n : f(\vec{x}) = \alpha\}$.*

(b) Express the linear program in the following form:

$$\max_{\vec{x} \in \mathbb{R}^n} \quad \vec{c}^\top \vec{x} \tag{14}$$

$$\text{s.t.} \quad A\vec{x} \leq \vec{b} \tag{15}$$

Specify the values of $\vec{x}$, $\vec{c}$, $A$, and $\vec{b}$.

(c) Express the linear program in the following form:

$$\min_{\vec{y} \in \mathbb{R}^n} \quad \vec{c}^\top \vec{y} \tag{16}$$

$$\text{s.t.} \quad A\vec{y} = \vec{b} \tag{17}$$

$$\vec{y} \geq 0 \tag{18}$$

Specify the values of $\vec{y}$, $\vec{c}$, $A$, and $\vec{b}$.

*HINT: Consider adding additional slack variables to the optimization problem.*

(d) List the extreme points or vertices of the feasible region of the linear program given by equations (8)–(13).

(e) Find the optimal value $p^\star$ and the optimal point $\vec{x}^\star = (x_1^\star, x_2^\star)$ of the linear program given by equations (8)–(13).

*HINT: Recall that for a linear program with a bounded feasible region, at least one optimal point is a vertex of the feasible region.*

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**5. Gordan's Lemma**

Let $A \in \mathbb{R}^{m \times n}$. Show that exactly one of the following is true:

(i) There exists $u \in \mathbb{R}^n$ such that $Au < 0$; or

(ii) There exists a nonzero $v \in \mathbb{R}^m$ with $v \geq 0$ and $A^T v = 0$.

*Hint: Try to use Farka's lemma with the matrix* $\begin{bmatrix} A^T \\ 1^T \end{bmatrix}$ *and the vector* $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

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**6. Homework Process**

With whom did you work on this homework? List the names and SIDs of your group members.

*NOTE*: If you didn't work with anyone, you can put "none" as your answer.

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